

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

MID Semester Examination
Course Number: PHY 4241
Course Title: Physics II

Summer Semester: 2023-2024

Full Marks: 120

Time: 2 Hours

Please answer according to the order of the questions. Answer all 4 (FOUR) questions. The symbols have their usual meanings. Marks of each question and the corresponding CO and PO are written in the brackets.

1. a) Define "free" electron in a metal. Electrical charge flowing through an imaginary surface of a copper wire is 0.16 Coul in a time of 64×10^{-3} sec, determine the current in amperes. How long it will take for 4×10^6 electrons to pass through the imaginary surface of Cu if the current is 5×10^{-3} A? (10)
(CO1)
(PO1)
- b) Name some of the active and passive elements. Draw a circuit diagram containing the active and passive elements which will represent an oscillatory circuit with dissipation of energy. How is the energy dissipated? Does it resemble a known mechanical system? (10)
(CO2)
(PO1)
- c) Discuss active and passive elements in an electrical circuit. Distinguish electrical resistance and resistivity. Right down Ohms law in scalar and vector form. How is the resistance of a material being affected by temperature? (10)
(CO2)
(PO1)
2. a) State Voltage divider rule. Draw the current-voltage diagrams of some Ohmic and non-Ohmic devices with specific reference to Thermistor. What does the negative slope mean in this device? (7)
(CO1.)
(PO1)
- b) Using a simple series circuit, derive an equation for the voltage divider rule. (8)
- c) For the series circuit (Fig. 1) (i) determine the total resistance R_T . (ii) Calculate the source current I_s (iii) Determine the voltage across each resistor (iv) Find the power supplied by the battery (v) Determine the power dissipated by each resistor (vi) Check if the total power supplied equals the total power dissipated (15)
(CO3)
(PO1)

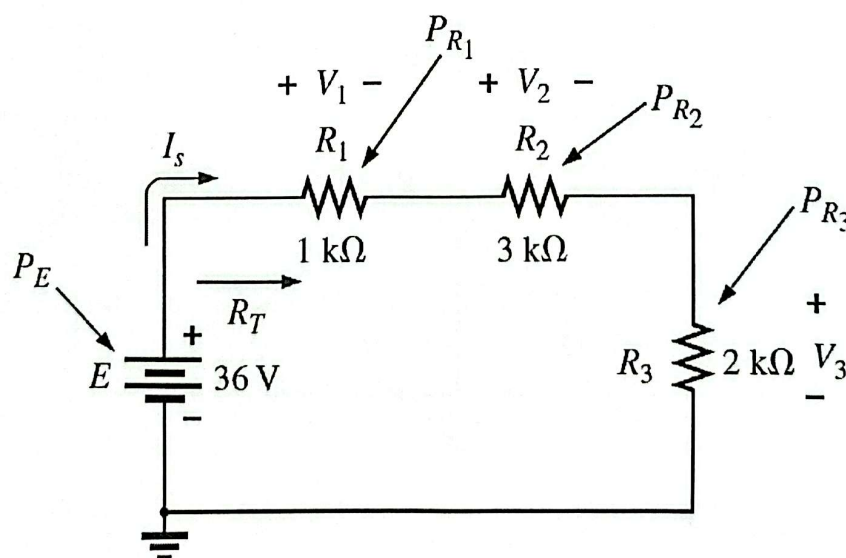


Figure 1: A series circuit for Q 2c

3. a) In a series parallel network what is meant by Reduce and Return approach?

(7)
(CO2)
(PO1)

b) Give an example showing the steps to reduce a circuit

(8)

c) For the parallel network (Fig.2) (i) Find the total resistance (ii) Calculate the source current (iii) Determine the current through each branch

(15)
(CO2)
(PO1)

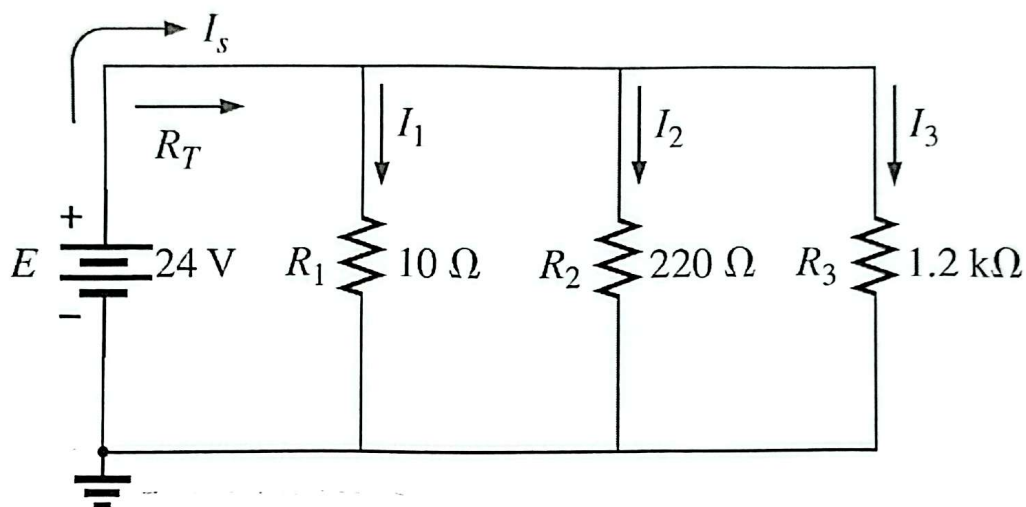


Figure 2: A parallel network for Q 3c

4. a) State and explain Superposition theorem.

(7)
(CO2)
(PO1)

b) Describe how would you apply the Superposition theorem to a circuit with two active devices such as voltage source and a current source.

(8)
(CO2)
(PO1)

c) Using the superposition theorem, determine the current through the 12 Ohm resistor in the circuit below (Fig. 3).

(15)

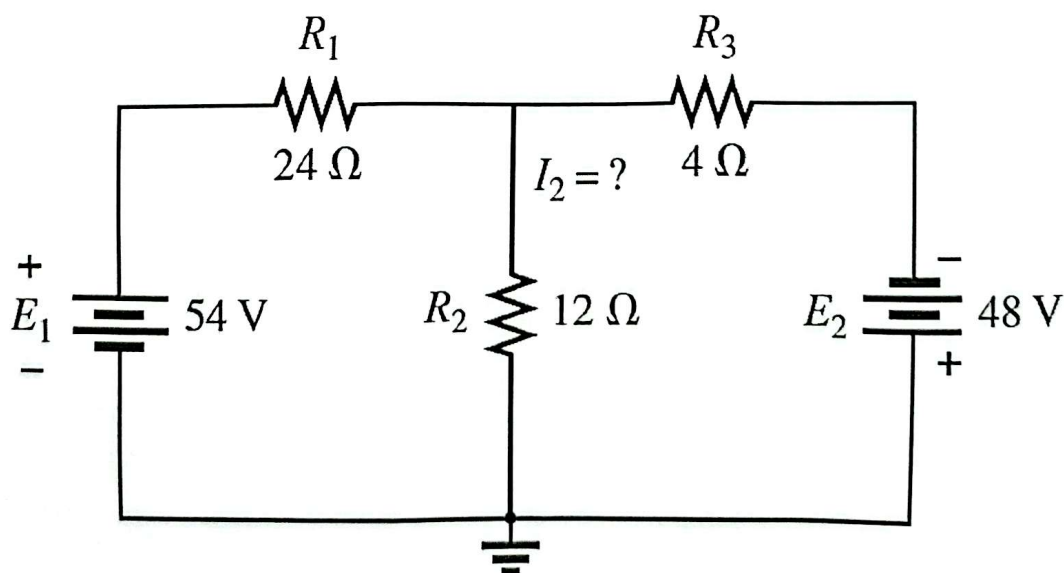


Figure 3: A sample circuit Q 4c

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 2 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 120

CSE 4203: Discrete Mathematics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Using the principle of Mathematical Induction, show that —
 - i. The closed formula for sequence $\{a_n\}$, defined by the recurrence relation, $a_n = 9.a_{n-1}$, is valid considering $n \in \mathbb{Z}^+$ for $n \geq 1$. 12
(CO2)
(PO1)
 - ii. $\forall n \in \mathbb{Z}^+, 9 \mid (10^n - 1)$. 12
(CO1)
(PO1)
- b) Simplify the following expressions — 3 × 2
(CO1)
(PO1)
 - i. $(p \vee (p \wedge q) \vee (p \wedge q \wedge \neg r)) \wedge ((p \wedge r \wedge t) \vee t)$
 - ii. $(p \wedge (\neg r \vee q \vee \neg q)) \vee ((r \vee t \vee \neg r) \wedge \neg q)$
2. a) **P1:** *All the policemen on this beat eat with our cook.*
P2: *No man with long hair can fail to be a poet.*
P3: *Amos Judd has never been in prison.*
P4: *Our cooks cousins all love cold mutton.*
P5: *None but policemen on this beat are poets.*
P6: *None but her cousins ever eat with the cook.*
P7: *Men with short hair have all been to prison.*
 Considering the premises **P1** through **P7** mentioned above in a Universal Domain, answer the following —
 - i. Derive the logical expressions of the premises **P1** through **P7**. 7
(CO2)
(PO1)
 - ii. Instantiate the predicates with the variable a , where " a : Amos Judd, who is a man", and state what facts can/cannot be logically derived about Mr. Amos Judd. 13
(CO3)
(PO2)
- b) Use rules of inference to determine if the following arguments are logically correct — 10
(CO1)
(PO1)
 - P1:** *There are men who are soldiers.*
 - P2:** *All soldiers are strong.*
 - P3:** *All soldiers are brave.*
 - Conclusion:** *Therefore, some strong men are brave.*

3. a) Prove the statement, "For any integer n, m , if n is even and m is odd then $2n + 5m$ is odd." 5
- b) If $m \in \mathbb{Z}^+$, $a \equiv b \pmod{m}$, and $c \equiv d \pmod{m}$ then show that 5 × 2
- $a + c = b +_m d$
 - $a \cdot c = b \cdot_m d$
- c) Suppose that a person deposits \$10,000 in a savings account at a bank yielding 11% per year with interest compounded annually. Based on this information, answer the following —
- Derive the recurrence relation and the corresponding closed formula for calculating the interest for the n^{th} year. 4 × 2
 - How much will be in the account after 30 years? 2
- d) Suppose there are signs on the doors to two rooms. The sign on the first door reads "*In this room there is a lady, and in the other one there is a tiger*"; and the sign on the second door reads "*In one of these rooms, there is a lady, and in one of them there is a tiger*." Suppose that you know that one of these signs is true and the other is false. Behind which door is the lady? 5
4. a) Consider the two functions, $f(x) = \frac{2x+3}{x+2}$ and $g(x) = \frac{3x+2}{x-3}$, which follows the condition, $\mathbb{Z}^+ \rightarrow \mathbb{Z}^+$. Compute the following —
- $f \circ g(x)$ 3
 - $g \circ f(x)$ 3
 - Determine if $f^{-1}(x)$ exists. If not, state any domain restrictions, if necessary, to ensure that $f(x)$ is a bijection and find $f^{-1}(x)$ based on the restriction you stated. 4
- b) Prove that if x is a real number, then $\lfloor 2x \rfloor = \lfloor x \rfloor + \left\lfloor x + \frac{1}{2} \right\rfloor$. 8
- c) Multiply the two binary numbers, $(101)_2$ and $(110)_2$. Apply the binary multiplication and addition algorithms as necessary with step-by-step illustration. 6 + 6

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DEPARTMENT OF NATURAL SCIENCES

Mid Semester Examination
 Course Code: **MATH 4241**
 Course Title: **Integral Calculus and Differential Equations**

Summer Semester: 2023-2024
 Full Marks: 160
 Time: 2 Hours

Programmable calculators are not allowed. Do not write anything on the question paper. Answer all 4 (four) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. (a) (i). Use the concept of sigma notation with x_i^* as the midpoint of each subinterval to find the area under the parabola $f(x) = 9 - x^2$ and over the interval $[0, 3]$. [12]
(CO1)
(PO1)
 - (ii). Write down the statement of Mean Value Theorem for Integrals (MVTI). Find all values of x^* for $f(x) = x^2 + x$ on the interval $[-12, 0]$ that satisfy the MVTI, and explain what these numbers represent.
 - (b) Write down the statement of Fundamental Theorem of Calculus. Hence, evaluate the following definite integrals: [16]
(CO2)
(PO1)
 - (i). $\int_0^{\ln 5} e^x (1 - 4e^x) dx$
 - (ii). $\int_0^1 \frac{dx}{(1+x)\sqrt{1+2x-x^2}}$
 - (iii). $\int_0^{\pi/2} \frac{dx}{4+5\sin x}$
 - (c) (i). Obtain a reduction formula for $\int \sec^n x dx$. [12]
(CO3)
 - (ii). If $I_n = \int_0^{\pi/4} \tan^n x dx$, then prove that $I_n + I_{n-2} = \frac{1}{n-1}$. Hence, evaluate I_7 . [12]
(PO1)
2. (a) (i). Define gamma and beta functions. Show that $\Gamma(n+1) = n\Gamma(n) = n!$. [20]
(CO1)
 - (ii). Show that $\int_0^{\pi/2} \sin^5 \theta \cos^4 \theta d\theta = \frac{8}{315}$. [20]
(PO1)

(iii). Prove that $\int_0^1 x^m \left(\ln \frac{1}{x} \right)^n dx = \frac{\Gamma(n+1)}{(m+1)^{n+1}}.$

(b) (i). Find and sketch the area of the region that is enclosed between the curves $y = x^2$ and $y = x + 6$. [12]
(CO2)

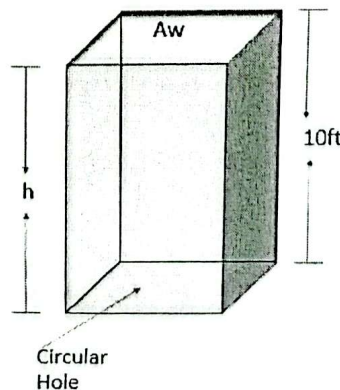
(ii). Find the volume of the solid generated when the region enclosed by the curves $x = y^2$ and $x = y + 2$ is revolved about the y -axis. (PO1)

(c) Use cylindrical shells to find the volume of the solid generated when the region that is enclosed by $y = \cos(x^2)$, $x = 0$, $x = \frac{1}{2}\sqrt{\pi}$, $y = 0$, is revolved about the y -axis. [8]
(CO4)
(PO1)

3. (a) Verify that the piecewise-defined function $y = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ is a solution of the differentiable equation $xy' - 2y = 0$ on $(-\infty, \infty)$. [12]
(CO1)
(PO1)

(b) Verify that the family of functions $y = e^{-x^2} \int_0^x e^{t^2} dt + c_1 e^{-x^2}$ is a solution of the ODE $\frac{dy}{dx} + 2xy = 1$. [13]
(CO2)
(PO1)

(c) Suppose water is leaking from a tank through a circular hole of area A_h at its bottom. When water leaks through a hole, friction and concentration of the stream near the hole reduce the volume of water leaving the tank per second to $cA_h\sqrt{2gh}$, where c ($0 < c < 1$) is an empirical constant. Determine a differential equation for the height h of water at time t for the cubical tank shown in figure. The radius of the hole is 2 in and $g = 32\text{ft/s}^2$. [15]
(CO3)
(PO1)



4. (a) Solve the given ODE (Any Three)

(i). $\frac{dy}{dx} = \frac{xy + 2y - x - 2}{xy - 3y + x - 3}$.

[24]

(CO3)

(PO3)

(ii). $y' + (\tan x)y = \cos^2 x, y(0) = -1$.

(iii). $\left(2y - \frac{1}{x} + \cos 3x\right) \frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$.

(iv). $(y^2 + xy)dx - x^2 dy = 0$.

(b) The radioactive isotope of lead Pb-209 decays at a rate proportional to the amount present at time t and has a half-life of 3.3 hours. If 1 gram of this isotope is present initially how long will it take for 90% of the lead to decay?

[16]

(CO4)

(PO4)

The End

Program: B.Sc C.S.E. 2nd Semester
Semester: Summer

Date: July 2, 2025 (Wednesday)
Time: 10:00 AM – 12:00 PM

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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DEPARTMENT OF TECHNICAL AND VOCATIONAL EDUCATION (TVE)

Examination: Mid-Semester Examination
Course Number: Hum 4241
Course Title: Islamic History, Science and Culture

Academic Year: 2023 – 2024
Full Marks: 80
Duration: 2.00 hours.

There are 4 (Four) questions. Answer all of them. The symbols have their usual meaning.

1. a) Explain the Evolution of Islamic Historiography, by briefly mentioning the contributions of popular Classic Muslim Historians- *Ibn Hisham, Ibn Qutaiba and Tabari.* 10 CO1 PO9/PO10
b) Why Ibn Khaldun's Contributions as a Historian praiseworthy? What do you know about '*Muqaddima ibn Khaldun*' and the theory of *Group Mind* and the *Philosophy of Nation State*? 10 CO2 PO10
2. a) Draw the Significance of the Hudaibiah Treaty as a great victory of Islam according to the Muslim Historians '*zuhri*' and '*Ibne Hisham*'s observations. 10 CO2 PO10
b) Evaluate the Significance of the *Madina Charter* according to the great historian William Muir's Book- "*A Life of Mahomet and History of Islam to the Era of the Hegira*". 10 CO4 PO8/PO12
3. a) Describe Prophet Muhammad's (PBUH) provincial reorganization (key provinces: Madinah, Makkah, etc.), and explain the roles of Wali, 'Amil, and Qāḍī in the decentralized yet supervised system for efficient governance. 10 CO3 PO6/PO7
b) Discuss the contemporary relevance of the Last Sermon of the Prophet (PBUH) in today's world. How can its universal values guide modern society? 10 CO4 PO8/PO12
4. **Write short notes on the following:** 20 CO1 PO9/PO10
 - (i) AD, BC, Sacred Months.
 - (ii) Farewell Pilgrimage.
 - (iii) Aws-Khazraj, Banu Nadir-Banu Qurayza.
 - (iv) "Live and Let Live others".
 - (v) Mu'akhāt (brotherhood).

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MID SEMESTER EXAMINATION
DURATION: 2 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 120

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.
 Ensure all necessary intermediate steps are shown to support the correctness of your answer.

Use a pencil for drawing diagrams, as it allows for easier corrections.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Similar to the concept of BCD (Binary-Coded Decimal), you are asked to design another binary code for base-12 numbers using the 7321 weighted code, which is often used to encode non-decimal digits in binary form for error detection or custom encoding schemes. 12 +
2 +
3 + 3
(CO2)
(PO2)
 - i. Construct a complete table of the 7321 weighted code for all base-12 digits
 - ii. Using the table, you constructed in Question 1.a.i), encode the base-12 number $B4A9$
 - iii. Is this weighted code a self-complementary code? Justify your answer.
 - iv. Does $(B4A9)_{16}$ have the same representation in the 7321 weighted code as the one obtained in Question 1.a.ii)? Justify your answer.
- b) Simplify the following Boolean expressions to a minimum number of literals following the cut-and-try method. 5 + 5
(CO1)
(PO1)
 - i. $(p + q)'(p' + q')$
 - ii. $(x'y' + z)' + z + xy + wz$
2. a) You are tasked to design a digital circuit to perform arithmetic operations on two unsigned decimal numbers, where each number consists of 3 BCD (Binary-Coded Decimal) digits. 15 +
5 + 5
(CO1)
(PO1)
 - i. Construct a block diagram to add two such 3-BCD-digit numbers using 4-bit binary adders labeling all blocks clearly. You may use a two-page span to fully illustrate this operation and make it readable.
 - ii. What is the range of the sum in decimal for this scenario? Justify your answer.
 - iii. What modifications would be required in your design from Question 2.a.i) to enable subtraction of such numbers instead of addition?
- b) Design a 4-bit binary-to-Gray code converter by following standard combinational circuit design procedure. Showing all necessary intermediate steps, draw the final logic diagram using universal gates. 10
(CO2)
(PO2)
3. In a digital communication system, the sender and receiver exchange 5-bit data messages. To ensure reliability during transmission, an error detection code is additionally used to identify any possible errors in the data.
 - a) Considering a simple and commonly used approach, design the circuit diagram for this system using only XOR or XNOR gates. Clearly indicate how this code is generated and checked considering all possible scenarios. 20
(CO2)
(PO2)
 - b) Discuss any limitation(s) of the error detection approach used in Question 3.a. Justify your answer with appropriate examples. 5
(CO3)
(PO2)

4. a) Using a Karnaugh Map, simplify the Boolean function F , incorporating the given don't care conditions D to obtain the minimal expression. Present the simplified expressions in both (i) NOR-OR and (ii) NAND-AND logic forms. 15
(CO2)
(PO2)

$$F(w, x, y, z) = wxy + w'xyz' + w'xy'z$$

$$D(w, x, y, z) = x'z + w'xz' + wx'z'$$

- b) Using Quine-McClusky (tabular) method, find the simplest product-of-sums format of the function $F = fg$ showing all intermediate steps, where: 15
(CO2)
(PO2)

$$f(a, b, c, d) = abc' + c'd + a'cd' + b'cd'$$

$$g(a, b, c, d) = (a + b + c' + d')(b' + c' + d)(a' + c + d')$$